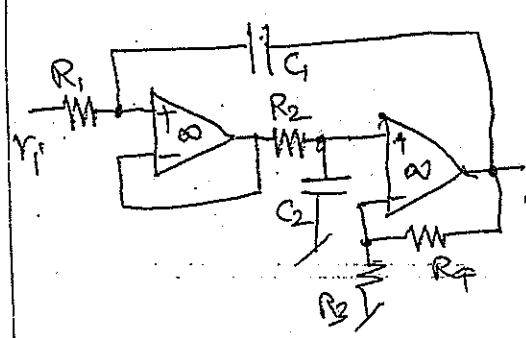
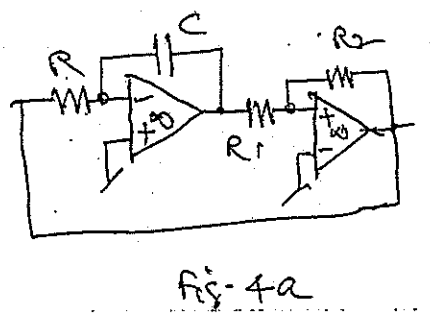
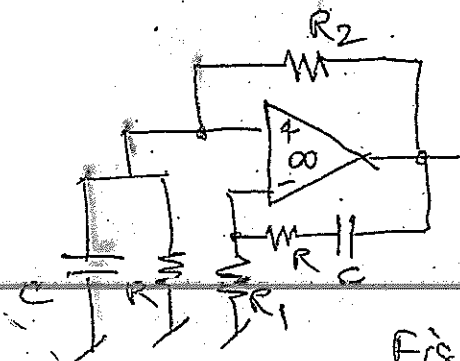


3c	From the general circuit diagram of the Antoniou's GIC derive the case of grounded inductor realisation and comment how its floating version can be made?	4	CO3
Q4	Attempt any 2 parts of the following		
4a	For the circuit given in Fig.4a discuss its functionality and derive expression for the wave generated.	4	CO4
4b	Draw an OPAMP-based monostable multivibrator and discuss its operation with necessary derivations and waveforms.	4	CO4
4c	Draw a neat internal diagram of 555 timer IC and, additionally, with appropriate external circuit elements explain its astable operation. Also derive expressions for T_{ON} and T_{OFF} .	4	CO4
	 Fig-3b  Fig-4a		
Q 5	Attempt any 2 parts of the following		
5a	Draw the block diagram of a Voltage-Shunt Feedback and deduce the expressions for R_{if} and R_{of} of the feedback amplifier in relation to R_i and R_o of the Amplifier without feedback.	4	CO5
5b	For the circuit given in Fig.5b analyse and test for its possibility of being an oscillator	4	CO5
5c	Draw the circuit diagram of a BJT-based Class-B push-pull configuration and derive expression for its efficiency, also discuss the harmonic distortion aspect of the same.	4	CO5
	 Fig-5b		